

Preparation of Antibacterial Nanocomposites of Zinc Oxide-Doped Graphene Reinforced Polypropylene with High Comprehensive Properties

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Polypropylene (PP) is one of the most widely used polymers in many areas. But it also has some drawbacks. In this work, a novel nanomaterial — zinc oxide-doped graphene (G-nZnO) — was melt-blended with PP to prepare an antibacterial nanocomposite. It was found that the

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mechanical properties, thermal stability and barrier properties of nanocomposites reached the highest value when the content of G-nZnO was 0.2 phr. The tensile properties of nanocomposites were 36% higher than those of pure PP, and the barrier properties were increased by 113%. The results of antibacterial tests showed that the antibacterial activity of nanocomposites increased with the G-nZnO content. Compared with previous studies on PP/graphene and different PP composites, this study that examined new nanocomposites was better in terms of improving various properties of PP. This kind of composite materials with excellent comprehensive performance has great potential for application in the fields of packaging and functional materials.

Keywords: Polypropylene; zinc oxide-doped graphene (G-nZnO); barrier property; antibacterial property; comprehensive performance.

1. Introduction

Polyolefins have been widely used in many areas.¹ Polypropylene (PP) is one of the most widely used polymers in textile and plastic industries.^{2,3} Over the years, the consumption rate of PP in various industries has been increasing.⁴ However, its applications are still limited because it lacks some properties. For example, PP cannot be used in the fields of medicine and health because it possesses no antibacterial abilities;^{5–7} it cannot be applied to aerospace because of its low stiffness.⁸ Therefore, it is extremely important to enhance certain properties of PP.

Many studies have shown that adding micro-materials or nanomaterials to polymer matrices can significantly improve the performance of raw materials and can also give them new properties.^{1,9–14} Radheshkumar *et al.*¹⁵ described that PP/silver composites with antibacterial properties have broad application prospects in antibacterial ophthalmic vials, packaging materials and multifilament yarns. Kada *et al.*¹⁶ recorded that carbon fiber/PP composites had been used in the automotive and aerospace industries in the last decade because carbon fibers improved electrical conductivity and mechanical stiffness. Hence, reinforcing PP or endowing it with some properties can broaden its applications.

To reinforce certain properties of PP, it has been filled with fillers that include zinc oxide,^{17–19} montmorillonite,^{8,20–22} nanoperlite,^{23,24} cellulose nanofiber^{25,26} and others.^{27–30} Graphene (G) is preferred due to its high conductivity, mechanical property and surface area.^{31–36}

In fact, graphene as a filler can make materials with better barrier properties, thermal stability, mechanical properties and electrical conductivity.^{37,38} Wang *et al.*³⁹ prepared graphene/poly(vinyl alcohol) nanocomposites with significantly enhanced mechanical

and barrier properties compared with pure PVA. Chaharmahali *et al.*⁴⁰ added graphene to bagasse/PP composites, which were melt-compounded using a twin-screw extruder, and the obtained composites had improved mechanical properties and thermal stability.

At present, there are many studies about PP/graphene composites. Song *et al.*³⁸ dispersed graphene evenly in PP, and the resultant nanocomposite exhibited better mechanical property and thermal stability than pure PP. Mistretta *et al.*⁴¹ combined graphene nanosheets with PP to improve the photooxidation performance of PP. There are also many studies about the electrical conductivity of PP.^{42–44} For example, You *et al.*⁴⁵ used polymethylmethacrylate to bridge graphene and the nonpolar PP matrix together; the storage modulus and conductivity of nanocomposites were improved in the in-plane direction. Milani *et al.*⁴⁶ prepared isotactic PP/graphene nanocomposites through in situ polymerization, where the electrical conductivity increased by a factor of 108 compared with pure isotactic PP.

Zinc oxide is regarded as a good antibacterial agent, which has been widely used in the field of packaging materials.^{47,48} But the role of zinc oxide as a filler is far more than that. Lepot *et al.*⁴⁹ found that zinc oxide could improve the tensile and oxygen barrier properties of biaxially oriented PP films. Marra *et al.*⁵⁰ found that combining zinc oxide with polylactic acid resulted in a bio-composite membrane with not only high antibacterial properties but also good mechanical properties and barrier properties for carbon dioxide and oxygen had also improved slightly.

In summary, there are a lot of research on graphene and zinc oxide applied to PP; however, research on these two nano-fillers that deal with improvement in the comprehensive performance of

PP has not yet been reported. Therefore, excellent properties of graphene and zinc oxide nanosheet were combined to produce a new type of nanomaterial in the form of zinc oxide-doped graphene (G-nZnO) by wet dispersion synthesis method, then PP was melt-blended with it to produce PP/G-nZnO nanocomposites with excellent properties. Their mechanical, barrier and antibacterial properties were analyzed.

2. Experimental

2.1. Materials and Synthesis of G-nZnO

PP (LH503) was produced by LCY Chemical Corporation (Kaohsiung, Taiwan). G-nZnO in this study was similar to previous study.¹² ZnO nanosheet (nZnO, the thickness is 20 nm and the diameter is 100–300 nm, see Fig. S1) and graphene was obtained from Zehan Co., Ltd. (Taipei, Taiwan), nZnO was capped on the surface of the graphene by using the wetting dispersion method. The ratio of graphene to nZnO was 1:2 (Table S1 and Fig. S2). The graphene was placed in an ethanol solution ultrasonic bath (FS-1088N Ultrasonic processor, SXSONIC Co., Ltd., Shanghai, China) for 1 h. Subsequently, nZnO was poured into the graphene-ethanol solution in an ultrasonic bath for 2 h to avoid aggregation. This solution was passed through a filter to remove the ethanol; subsequently, the mixture was placed in an oven at 60°C for 2 h to evaporate the residual ethanol, then adjust the oven temperature to 150°C for 1 h. Finally, the G-nZnO was obtained. The images taken from SEM and TEM are shown in Fig. S3.

2.2. Preparation of PP/G-nZnO nanocomposites

PP was melt-blended with 0.1 phr, 0.2 phr, or 0.3 phr G-nZnO by using a HAAKE mixer (Thermo Fisher Scientific, Germany) at a speed of 100 rpm and at 180°C for 4 min. Gray PP/(G-nZnO)^x nanocomposites with a uniform color was obtained. They were first vacuum-dried at 80°C for 8 h, then hot-pressed at 180°C in preparation for subsequent tests. The synthesis of G-nZnO and preparation of PP/G-nZnO nanocomposites are shown in Scheme 1.

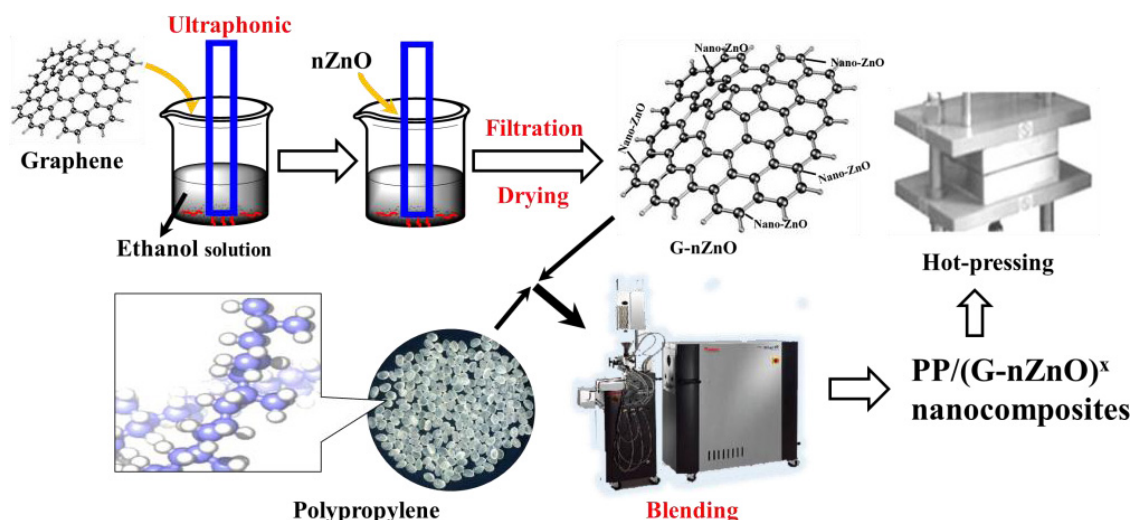
2.3. Characterization PP and PP/G-nZnO nanocomposites

2.3.1. Characterization techniques

Scanning electron microscopy (SEM, TESCAN, Czech) was used to observe the defect of PP and its nanocomposites. Photos were taken at the same magnification. The dispersed nanofillers in nanocomposites were also obtained using an energy dispersive spectrometer (EDS, BRUKER, German).

Mechanical tests were carried out on a tensile testing machine (Furbs, Xiamen Furbusi Testing Equipment Co. Ltd., Xiamen, China) operated at a speed of 50 mm/min. Then, mechanical properties, including tensile strength (TS, σ , MPa) and elongation at break (EAB, ε , %), were obtained.

Differential scanning calorimetry (DSC) was conducted on a DSC-200 F3 calorimeter (Netzsch, Germany) that was operated under nitrogen (N₂)



Scheme 1. The synthesis of G-nZnO and preparation of PP/G-nZnO nanocomposites.

environment. Samples were heated to 300°C from room temperature at a rate of 10°C/min, and then cooled to the primary temperature at the same rate. Afterward, they were reheated to 300°C at the same rate. The first melting temperature (T_{m1}) and crystallization temperature (T_c) were obtained from the first heating and cooling curves, respectively. The second melting temperature (T_m) and crystallinity were obtained from the second heating curve.

Thermogravimetric analysis (TGA) was performed using TGA instruments (Model STA 409PC, Netzsch, Germany), which were operated under N₂ environment. Heating was done from room temperature to 750°C at a rate of 10°C/min. TGA and DTG (derivative of TGA) curves were obtained.

X-ray diffraction (XRD) patterns were obtained using Bruker-D2 HASER (Germany) operating on Cu-K α radiation at room temperature. Data were recorded every 0.2 s in the range 5–90°. The crystallization behavior of PP and its composites was obtained.

The water vapor permeation rate (WVPR) was evaluated according to ASTM E96 standard methods at 37.8°C and 90% relative humidity. WVPR (g/m².h) was calculated using the following equation:

$$WVTR = W_g/tA, \quad (1)$$

where W_g was the weight measured at different periods of time, t was the test time and A was the permeation area of samples.

The oxygen transmission rate (OTR) was determined based on the standard of ASTM D1434 at 23°C and 0% relative humidity. OTR (cm³cm/(cm²·s·Pa)) was measured from the following relationship:

$$OTR = Q/At(P_1 - P_2), \quad (2)$$

where Q was the amount of gas passing through the membrane, A was the area of contact between the gas and the membrane, t was the time and ($P_1 - P_2$) was the local pressure difference.

2.3.2. Antibacterial tests

The antibiotic activity of PP and its composites was determined through qualitative and quantitative tests against *E. coli*. The method used was described by Tsou et al.^{5,6} For the qualitative test, samples were placed in an agar petri dish with microbes, and then incubated at 37°C for 24 h. There was a clean

zone around the samples if inhibitory effects were reached. For the quantitative test, each sample was cultured with an *E. coli* suspension in a serum bottle, and it was shaken at a speed of 120 rpm/min at 37°C for 24 h. The sample antibacterial activity was gauged from the number of counted bacteria.

3. Results and Discussion

3.1. Characterization techniques

3.1.1. Morphology

Figure 1 presents fractured surfaces of PP and PP/G-nZnO. Relative to G-nZnO in PP/(G-nZnO)^{0.2} and PP/(G-nZnO)^{0.3}, G-nZnO in PP/(G-nZnO)^{0.1} was not exposed, because lower amounts of nanofillers could be covered well by the substrate. Meanwhile, PP and its nanocomposites all showed rough surfaces, which might be produced by the production process [Figs. 2(a)–2(d)]. When G-nZnO was 0.3 phr, G-nZnO appeared obviously, there was an obvious gap between the nanofiller and the PP substrate, which may be caused by slight agglomeration or incompatibility.

EDS was used to measure the content of Zn in PP and its nanocomposites, which could be demonstrated by the dispersion of G-nZnO [Figs. 2(a')–2(d')]. There was no Zn in pure PP. Nanofillers were dispersed uniformly in nanocomposites when the G-nZnO content was 0.1 phr and 0.2 phr. As expected, Zn increased with the G-nZnO content. When the nanofiller reached 0.3 phr, there was still no serious agglomeration. Therefore, nanofillers were relatively well-dispersed in PP.

3.1.2. Mechanical property

Figure 3 presents the data on the TS and EBA of PP and PP/G-nZnO nanocomposites. The TS of PP/G-nZnO nanocomposites was higher than that of pure PP, and the respective improvements for the nanocomposites are 30% and 36% at loadings of 0.2 phr and 0.3 phr G-nZnO, respectively. Meanwhile, the study by An et al.⁵¹ and Achaby et al.⁵² showed that mechanical properties greatly increased with the graphene concentration. According to a study by Zhao et al.,¹⁷ nano-ZnO could improve the EAB of materials. Therefore, the maximal reinforcement in EAB was 41%, which was achieved at 0.2 phr G-nZnO; then, there was a downturn due to the slight defect between nanofiller and PP in PP/

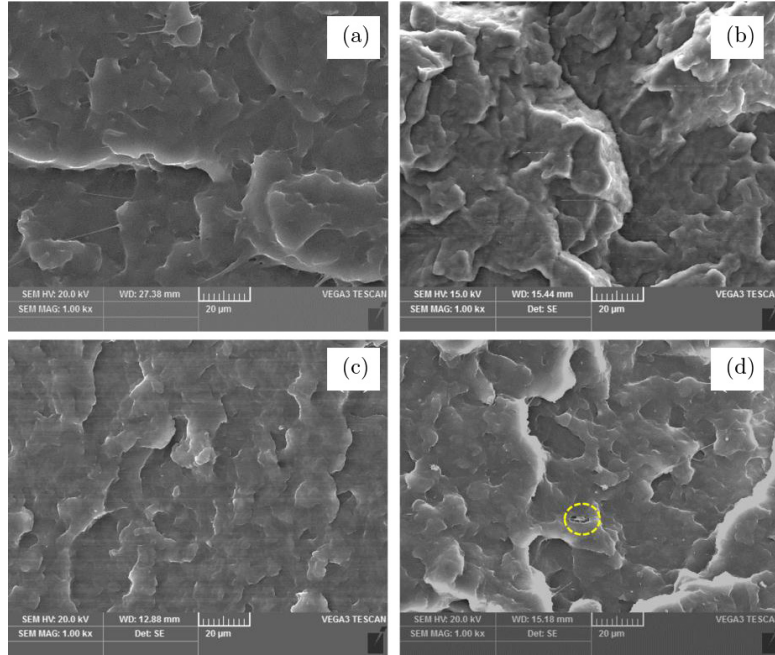


Fig. 1. SEM images of PP and its nanocomposite: (a) PP, (b) PP/(G-nZnO)^{0.1}, (c) PP/(G-nZnO)^{0.2} and (d) PP/(G-nZnO)^{0.3}

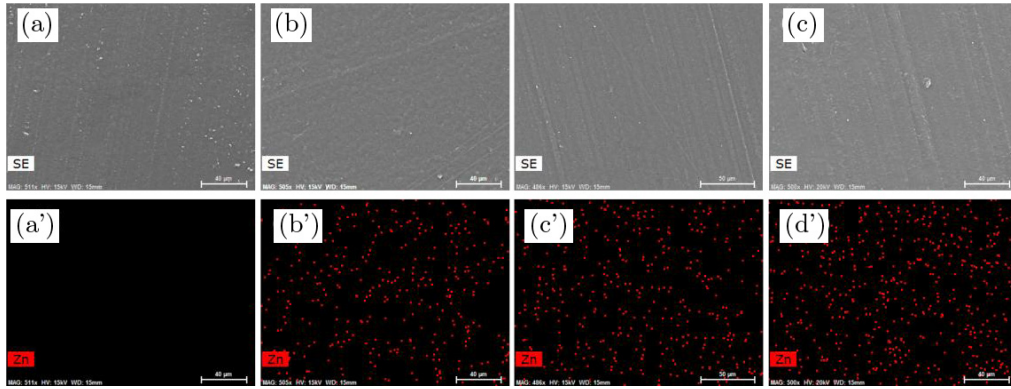


Fig. 2. Surface SEM images (a-d) and surface EDS images (a'-d') illustrating the distribution of Zn: (a, a') PP, (b, b') PP/(G-nZnO)^{0.1}, (c, c') PP/(G-nZnO)^{0.2} and (d, d') PP/(G-nZnO)^{0.3}

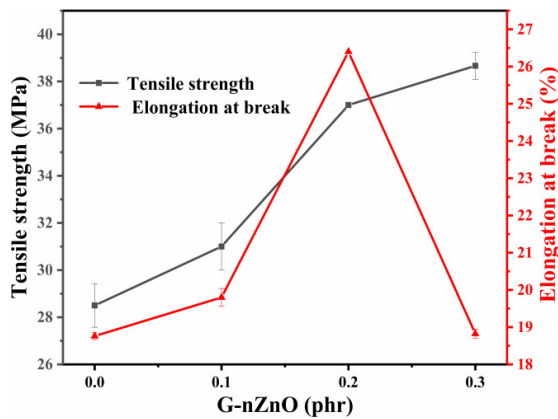


Fig. 3. Tensile strength and elongation at break of PP and its nanocomposites.

(G-nZnO)^{0.3} [Fig. 1(d)], resulting from the slight agglomeration of G-nZnO [Fig. 2(d)]. Overall, 0.2 phr G-nZnO was a more suitable amount to be added to PP, because the tensile strength was slightly increased when the content was increased from 0.2 to 0.3 phr, and EAB also went down at this range.

3.1.3. Differential scanning calorimetry

Figure 4 shows DSC curves for PP and its nanocomposite. Graphene and nanoZnO could play the role of nucleating agents because they both had high nucleation abilities.⁵³⁻⁵⁶ Hence, $T_{c2} - T_{c1}$ for

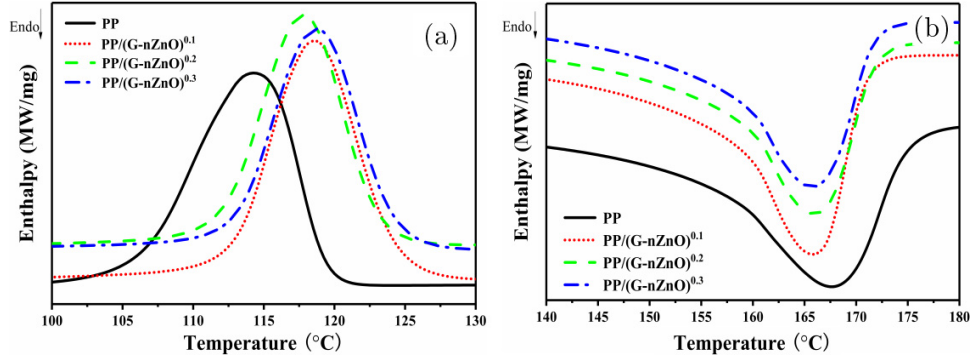


Fig. 4. DSC curves for PP and its nanocomposites: (a) first cooling curves and (b) second heating curves.

Table 1. DSC data for PP and its nanocomposites.

Sample	T_{c1} (°)	T_{c2} (°)	$T_{c2} - T_{c1}$ (°)	T_c (°)	T_m (°)	ΔH_m (J/g)	X_c (%)
PP	106.3	119.4	13.1	114.3	167.5	95.81	46
PP/(G-nZnO) ^{0.1}	112.9	124.3	11.4	118.6	165.7	90.99	44
PP/(G-nZnO) ^{0.2}	112.1	123.1	11	117.8	166.0	81.25	40
PP/(G-nZnO) ^{0.3}	113.1	124.2	11.1	118.7	165.8	81.86	40

PP/G-nZnO was lower than that for PP (Table 1). However, the overall T_c shifted to the right [Fig. 4(a)], with PP serving as reference, which means that G-nZnO could promote PP to melting crystallization.⁵⁰ Crystallinity (X_c) was evaluated from the following relationship:

$$X_c(\%) = \Delta H_m / (1 - \alpha) \Delta H_m^0 \times 100\%, \quad (3)$$

where α was the content of G-nZnO, ΔH_m was the measured melting enthalpy, and ΔH_m^0 was the enthalpy of 100% PP crystals (207.1 J/g).⁵⁷ On one hand, decreased X_c might be ascribed to the addition of G-nZnO, which promoted melt crystallization and increased the crystallization temperature that led to the shortening of crystallization time. On the other hand, G-nZnO promoted the refinement of

polymer crystals, which made PP crystals less complete; thus, crystallinity and T_m decreased.^{58–60} The T_{m1} also declined (Fig. S4). Therefore, G-nZnO could promote heterogeneous nucleation and increase the melting crystallization rate.¹⁴

3.1.4. Thermogravimetric analysis

TGA and DTG curves for PP and its nanocomposites are represented in Fig. 5, and these data are summarized in Table 2. It is obvious that pure PP showed weight losses at lower temperatures [Figs. 5(a) and 5(b)]. The T_d and fastest degradation temperature (T_d from DTG) of nanocomposites were higher than those of PP when G-nZnO was added. Meanwhile, when samples were at the same

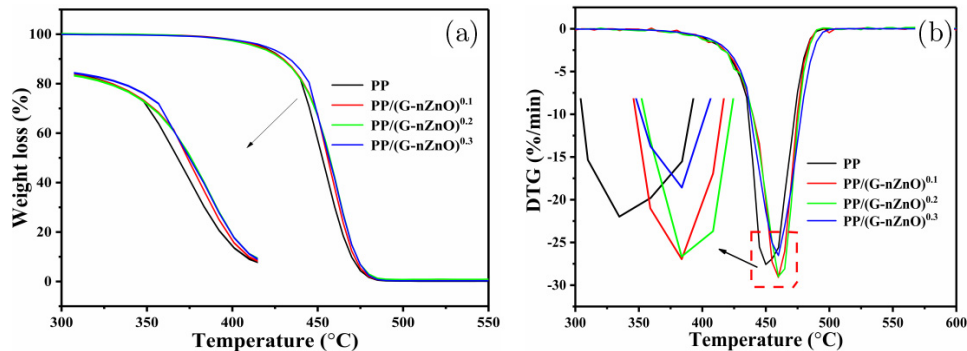


Fig. 5. (a) TGA and (b) derivative thermogravimetry for PP and its nanocomposites.

Table 2. Thermogravimetric data for PP and its nanocomposites.

Sample	$T_{5\%}$ (°)	$T_{10\%}$ (°)	T_d (°)	T_d from DTG
PP	414.9	429.9	435.0	451.3
PP/(G-nZnO) ^{0.1}	414.9	429.9	441.5	452.8
PP/(G-nZnO) ^{0.2}	414.9	429.9	440.9	461.5
PP/(G-nZnO) ^{0.3}	419.9	434.9	442.6	460.4

weight loss rates, the temperature of PP with added 0.3 phr G-nZnO was the highest, and the DTG peak was the highest when the G-nZnO content was 0.2 phr (Table 2). DTG curves showed temperatures at which weight loss rates were maximum, indicating that PP/G-nZnO nanocomposites required higher temperatures to thermally degrade. Therefore, G-nZnO could improve the thermal stability of PP.

3.1.5. X-ray diffraction

Figure 6 illustrates XRD patterns of PP and its nanocomposites. From the comparison of XRD curves for pure PP and PP/G-nZnO nanocomposites, it was found that new peaks did not appear for the case of PP/G-nZnO, indicating that G-nZnO was successfully dispersed in the PP matrix.⁵² However, peaks for PP/G-nZnO shifted slightly to the left, indicating that lattice parameters became larger, and the spacing between crystal planes increased, which may be due to the larger heteroatoms affecting, to some extent, the crystallization behavior of PP.³⁸ Hence, it was a readily obtainable result indicating that the presence of G-nZnO affected the crystallization behavior of the matrix. The results from DSC showed that G-nZnO would cause grain refinement, which further proved that G-nZnO would hinder the growth of PP crystals,

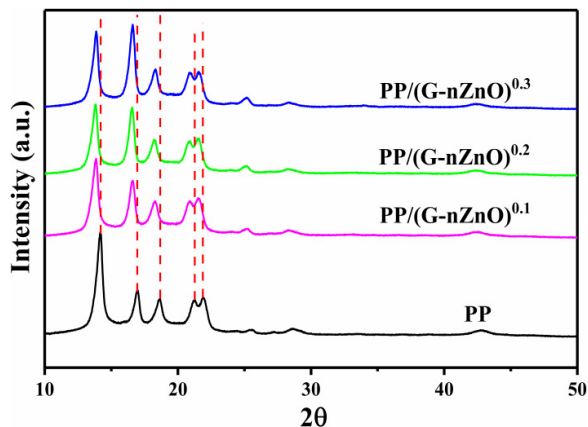


Fig. 6. XRD plots for PP and its nanocomposites.

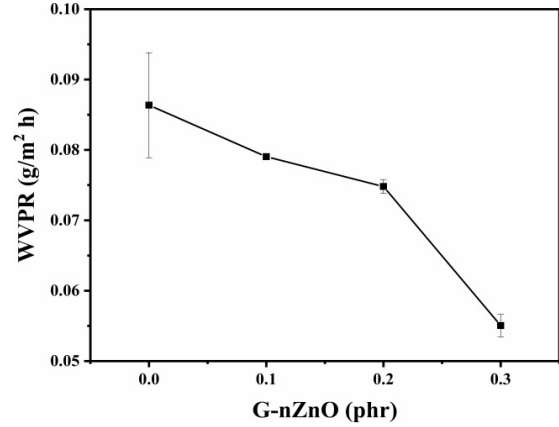


Fig. 7. WVPR for PP and its nanocomposites.

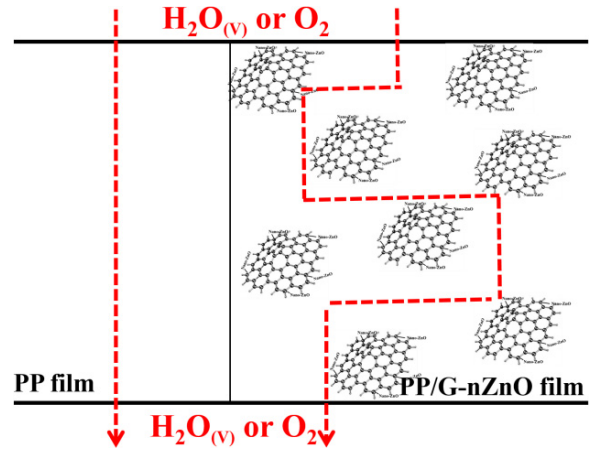


Fig. 8. Schematic diagram of water vapor (oxygen) permeation rate for PP and its nanocomposites.

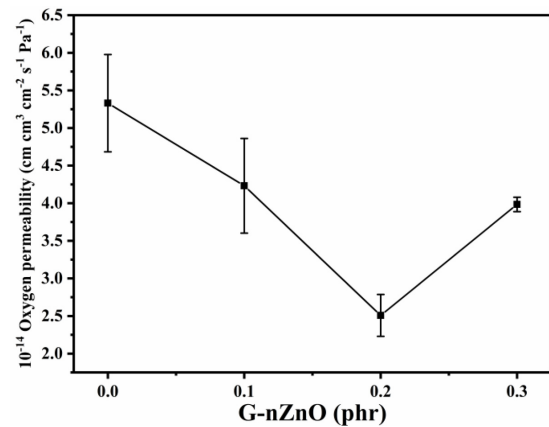


Fig. 9. OTR of PP and its nanocomposites.

resulting in larger spacing between the crystal planes. Therefore, both DSC and XRD results showed that G-nZnO did affect the crystallization behavior of PP.

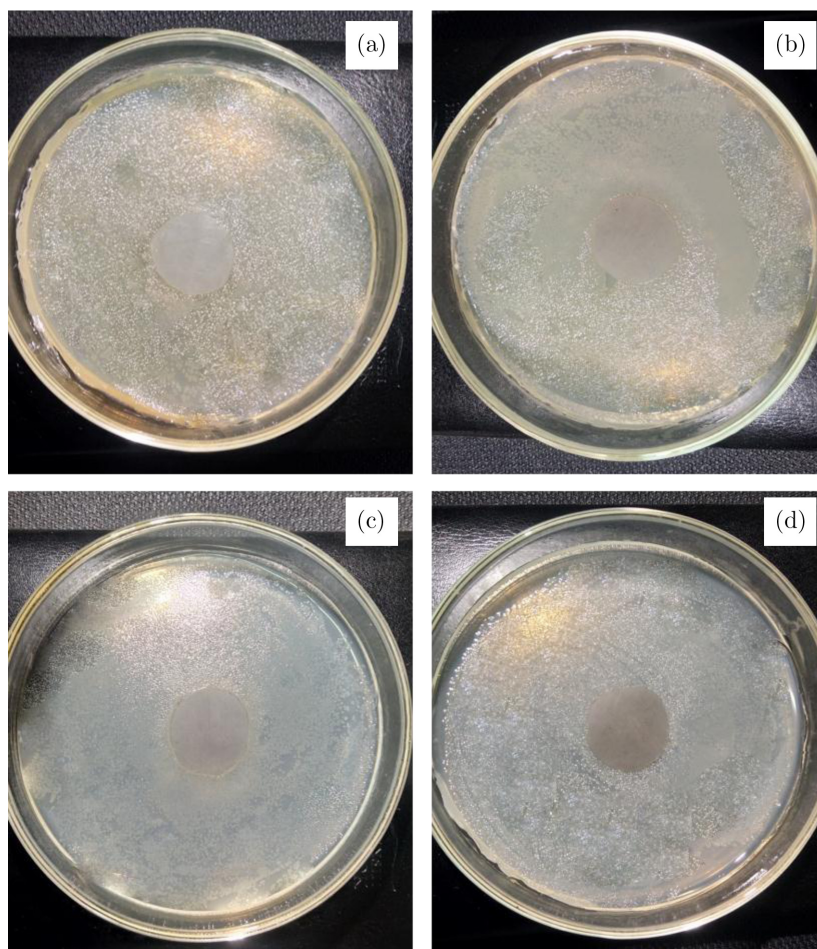


Fig. 10. Zones of inhibition: (a) PP, (b) PP/(G-nZnO)^{0.1}, (c) PP/(G-nZnO)^{0.2} and (d) PP/(G-nZnO)^{0.3}

3.1.6. Water vapor permeation

Figure 7 exhibits WVPR results for PP and its nanocomposites. WVTR for PP/G-nZnO nanocomposites was lower than that for pure PP. When the G-nZnO content was 0.3 phr, the nanocomposite had the best water vapor barrier property. The path of water vapor (or oxygen) through PP/G-nZnO nanocomposites was more complicated than through the pure sample (Fig. 8), so PP/G-nZnO nanocomposites had good barrier properties. At the same time, previous studies have shown that the addition of nanomaterials could improve the barrier properties of materials.^{10,61,62}

3.1.7. Oxygen transmission rate

Nanofillers can improve the barrier property of materials, because nanofillers act as physical barriers that delay the passage of oxygen through the polymeric matrix.⁶³ Figure 9 presents the results of

oxygen barrier tests on PP and its nanocomposites. It was obvious that the oxygen barrier property of PP/G-nZnO nanocomposites was better than that of PP. When the G-nZnO content was 0.1 phr, the error bar was larger due to less filler content and fewer physical obstacles. The maximal reinforcement on OTR was at 0.2 phr, and it then decreased.

Comprehensive results of tensile properties, TGA and OTR further indicated that the optimal additive content was reached when the G-nZnO content was 0.2 phr. At that condition, this new nanofiller

Table 3. Antibacterial rate for PP and its nanocomposites tested on *E. coli*.

Sample	(10 ⁶) CFU	Antibacterial rate (%)
Control	10	0
PP	10	0
PP/(G-nZnO) ^{0.1}	1	90
PP/(G-nZnO) ^{0.2}	0.2	98
PP/(G-nZnO) ^{0.3}	0.024	99.76

Table 4. Comparison of data on tensile and water vapor and oxygen barrier performance for various PP nanocomposites.

Sample	Mechanical property		Water vapor barrier performance		Oxygen barrier performance		Antibacterial property	Reference
	TS (MPa)	Promotion rate (%)	WVPR g/m ² ·h	Promotion rate (%)	10 ⁻¹⁴ cm ³ cm/(cm ² ·s·Pa)	Promotion rate (%)	Antibacterial rate (%)	
PP/G-nZnO	38.7	36	0.055	57	2.51	113	99.76	This work
PP/G	37.4	12	—	—	—	—	—	31
PP/FGS	35.5	18	—	—	—	—	—	64
PP/CN	31.5	14	—	—	5.14	67	—	65
DA-GO/iPP	—	—	—	—	2.68	177	—	66
PP/ZnO	—	—	—	—	5.54	14	—	67
PP-PEGC	—	—	—	—	7.44	43	—	68
PP/clay/ZnO	—	—	—	—	4.91	33	—	67

had sufficient content and was uniformly dispersed in the polymer substrate, so the comprehensive properties of PP were improved. According to the analysis results of SEM, it is known that agglomeration caused defects at a loading of 0.3 phr G-nZnO, and the oxygen molecules were very small, it was easy for them to pass through defective paths. But overall, G-nZnO could improve the oxygen barrier for PP.

3.2. Antibacterial tests

Figure 10 is the qualitative test results for PP and PP/G-nZnO nanocomposites tested against *Escherichia coli*. Nanocomposites and pure samples had similar results: there were no obvious zones of inhibition. This may be because G-nZnO was covered well by the substrate, and there was none exposed. Hence, nanofillers were not in direct contact with the bacteria, thereby the bacteriostatic circle was not obvious. However, from quantitative tests, the antibacterial activity of nanocomposites against *E. coli* (Table 3) increased with the G-nZnO content. The antibacterial rate was calculated using the following formula:

$$\begin{aligned} & \text{Antibacterial rate(\%)} \\ &= \text{CFU}_{\text{Control}} - \text{CFU}_{\text{Sample}} / \text{CFU}_{\text{Control}} \times 100\%. \end{aligned} \quad (4)$$

The best nanocomposite had an antibacterial rate of 99.76%. Therefore, antibacterial properties were conferred on PP by G-nZnO.

3.3. Comprehensive performance comparison

Table 4 lists comprehensive properties about tensile strength and barrier and antibacterial properties

obtained in this study, compared with those of PP-loaded graphene, nZnO and other nanomaterials reported in other studies. The results indicated that the tensile strength and improvements in other studies were lower than those in this study.^{31,64,65} As shown in Fig. 8, flaky G-nZnO might had better capacity to block or extend the path of oxygen molecules, which might be attributed to its large plane area. Lei *et al.*⁶⁶ also reported that functionalized graphene oxide could improve the oxygen resistance of PP, but the actual oxygen barrier ability was lower than that in this work, and the tensile property was not shown. Meanwhile, graphene and PP nanocomposites were seldom used in studies about water or oxygen resistance, so we compare them with other nanocomposites. It is shown from Table 4 that oxygen barrier performance and improvement rates of nanocomposites in this study were obviously superior to other nanocomposites.^{67,68} On the other hand, only in this study that nanocomposites were investigated for antibacterial properties. Except for the report by Atayev *et al.*,⁶⁷ whose samples may have excellent antibacterial properties due to the strong photocatalytic effect of nZnO, the results of other studies should have no effective antibacterial activity. In conclusion, the results in this work demonstrated the best improvement in comprehensive performance of PP nanocomposites.

4. Conclusion

Research on the influence of single graphene on the performance of polymers has been numerous, but G-nZnO has not been widely used in the field of polymers, especially general polymers, such as PP, and there is no study on the comprehensive

performance of these nanocomposites. This study is dedicated to evaluating the effect of G-nZnO on PP. The study found that G-nZnO enhanced the tensile strength, thermal stability and barrier properties of the matrix and gave PP antibacterial properties. Meanwhile, it also promoted the PP crystallization rate. When G-nZnO content was 0.2 phr, the mechanical properties, thermal stability and barrier properties of the nanocomposite were the best. A comprehensive comparison of previous studies and our study indicated that G-nZnO improved the performance of PP the most. This PP/G-nZnO nanocomposite has a broad market application prospect; it has potential to be used in food packaging materials that require high barrier performance and antibacterial properties, as well as in engineering materials that require strength, etc.

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References

1. B. Wang, D. Peng, R. Lv, B. Na, H. Liu and Z. Yu, *Compos. Sci. Technol.* **177**, 1 (2019).
2. Y.-H. Wen, M. R. De Guzman, X. Lin, J.-C. Chen, C. Gao, C.-S. Wu, Z.-H. Wang, D. Huang, J. Du, Y.-Q. Yu, W. Zhao, M.-C. Suen and C.-H. Tsou, *Nano Br. Rep. Rev.* **15**, 2050112 (2020).
3. M. Altan and H. Yildirim, *J. Mater. Sci. Technol.* **28**, 686 (2012).
4. M. Ajorloo, M. Fasihi, M. Ohshima and K. Taki, *Mater. Des.* **181**, 108068 (2019).
5. C.-H. Tsou, C.-S. Wu, W.-S. Hung, M. R. De Guzman, C. Gao, R.-Y. Wang, J. Chen, N. Wan, Y.-J. Peng and M.-C. Suen, *Polymer (Guildf)*. **160**, 265 (2019).
6. C.-H. Tsou, W.-H. Yao, W.-S. Hung, M.-C. Suen, M. De Guzman, J. Chen, C.-Y. Tsou, R. Y. Wang, J.-C. Chen and C.-S. Wu, *Ind. Eng. Chem. Res.* **57**, 2537 (2018).
7. W.-H. Yao, C.-H. Tsou and J.-C. Chen, *Fibres Text. East. Eur.* **24**, 117 (2016).
8. J. Gou, L. Zhang and C. Li, *Polym. Test.* **82**, 106236 (2019).
9. B. Yuan, G. Chen, Y. Zou, S. Shang, Y. Sun, B. Yu, S. He and X. Chen, *Polym. Adv. Technol.* **30**, 2153 (2019).
10. C.-H. Tsou, H.-T. Lee, W.-S. Hung, M. De Guzman, S.-T. Chen, M.-C. Suen and S. T. Wicaksono, *Polym. Bull.* **74**, 1121 (2017).
11. E. Gharoy Ahangar, M. H. Abbaspour-Fard, N. Shahtahmassebi and M. C. P. Maddahi, *J. Food Process. Preserv.* **39**, 1142 (2015).
12. C.-H. Tsou, W.-H. Yao, Y.-C. Lu, C.-Y. Tsou, C.-S. Wu, J. Chen, R. Wang, C. Su, W.-S. Hung, M. De Guzman and M.-C. Suen, *Polymers (Basel)*. **9**, 100 (2017).
13. C.-H. Tsou, H.-T. Lee, W.-S. Hung, C.-C. Wang, C.-C. Shu, M.-C. Suen and M. De Guzman, *Polymer (Guildf)*. **85**, 96 (2016).
14. C. H. Tsou, B. J. Kao, M. C. Suen, M. C. Yang, T. Y. Wu, C. Y. Tsou, J. C. Chen, W. H. Yao, C. K. Chu and X. M. Tuan, *Mater. Res. Innov.* **18**, S2 (2014).
15. C. Radheshkumar and H. Münstedt, *React. Funct. Polym.* **66**, 780 (2006).
16. D. Kada, A. Koubaa, G. Tabak, S. Migneault, B. Garnier and A. Boudenne, *Polym. Compos.* **39**, E664 (2018).
17. H. Zhao and R. K. Y. Li, *Polymer (Guildf)*. **47**, 3207 (2006).
18. J.-W. Zha, Q.-Q. Qin and Z.-M. Dang, *IEEE Trans. Dielectr. Electr. Insul.* **26**, 868 (2019).
19. J. Guo, G. Liu, Y. Guo, L. Tian, X. Bao, X. Zhang, B. Yang and J. Cui, *J. Polym. Res.* **26**, 19 (2019).
20. S. Zhu, J. Chen, Y. Zuo, H. Li and Y. Cao, *Appl. Clay Sci.* **52**, 171 (2011).
21. J. Ma, S. Zhang, Z. Qi, G. Li and Y. Hu, *J. Appl. Polym. Sci.* **83**, 1978 (2002).
22. Q. Zhang, Q. Fu, L. Jiang and Y. Lei, *Polym. Int.* **49**, 1561 (2000).
23. R. Sahraeian, S. M. Davachi and B. S. Heidari, *Compos. Part B Eng.* **162**, 103 (2019).
24. S. M. Davachi, B. S. Heidari, R. Sahraeian and A. Abbaspourrad, *Compos. B. Eng.* **175**, 107088 (2019).
25. L. Wang, K. Okada, Y. Hikima, M. Ohshima, T. Sekiguchi and H. Yano, *Polymers (Basel)*. **11**, 249 (2019).

26. Y. Peng, S. A. Gallegos, D. J. Gardner, Y. Han and Z. Cai, *Polym. Compos.* **37**, 728 (2016).
27. K. Zhao, S. Li, M. Huang, X. Shi, G. Zheng, C. Liu, K. Dai, C. Shen, R. Yin and J. Z. Guo, *Chem. Eng. J.* **358**, 924 (2019).
28. I. Rytöluoto, M. Niittymäki, K. Lahti, E. Saarimäki, T. Flyktman, M. Pääjärvi and M. Karttunen, *Proc. Nord. Insul. Symp.* **26**, 123 (2019).
29. B. Maillhot, S. Morlat, J.-L. Gardette, S. Boucard, J. Duchet and J.-F. Gérard, *Polym. Degrad. Stab.* **82**, 163 (2003).
30. S. Y. Yeo and S. H. Jeong, *Polym. Int.* **52**, 1053 (2003).
31. L. Burk, M. Walter, A. C. Asmacher, M. Gliem, M. Moseler and R. Mülhaupt, *Exp. Polym. Lett.* **13**, 286 (2019).
32. A. Allahbakhsh, F. Sharif and S. Mazinani, *Nano Br. Rep. Rev.* **8**, 1350045 (2013).
33. C. H. Tsou, Q. F. An, S. C. Lo, M. De Guzman, W. S. Hung, C. C. Hu, K. R. Lee and J. Y. Lai, *J. Memb. Sci.* **477**, 93 (2015).
34. R. Rong, H. Zhao, X. Gan, S. Chen and X. Quan, *Nano Br. Rep. Rev.* **12**, 1750008 (2016).
35. Z. Sun, Q. Fu, H. Li and K. Li, *Nano Br. Rep. Rev.* **9**, 1450066 (2014).
36. J. Z. Xu, Y. Y. Liang, H. D. Huang, G. J. Zhong, J. Lei, C. Chen and Z. M. Li, *J. Polym. Res.* **19**, 9975 (2012).
37. P. Enrique-Jimenez, S. Quiles-Díaz, H. J. Salavagione, J. P. Fernández-Blázquez, M. A. Monclús, R. G. de Villoria, M. A. Gómez-Fatou, F. Ania and A. Flores, *Compos. Sci. Technol.* **169**, 151 (2019).
38. P. Song, Z. Cao, Y. Cai, L. Zhao, Z. Fang and S. Fu, *Polymer (Guildf.)* **52**, 4001 (2011).
39. J. Wang, X. Wang, C. Xu, Z. Min and X. Shang, *Polym. Int.* **60**, 816 (2011).
40. M. Chaharmahali, Y. Hamzeh, G. Ebrahimi, A. Ashori and I. Ghasemi, *Polym. Bull.* **71**, 337 (2014).
41. M. C. Mistretta, L. Botta, A. D. Vinci, M. Ceraulo and F. P. La Mantia, *Polym. Degrad. Stab.* **160**, 35 (2019).
42. R. M. del Castillo, L. F. del Castillo, A. G. Calles and C. Vicente, *J. Mater. Chem. C* **6**, 7232 (2018).
43. P. Chammingkwan, K. Matsushita, T. Taniike and M. Terano, *Materials (Basel)* **9**, 240 (2016).
44. K. A. Imran, J. Lou and K. Shivakumar, *J. Appl. Polym. Sci.* **135**, 45833 (2018).
45. F. You, X. Li, L. Zhang, D. Wang, C. Shi and Z. Dang, *RSC Adv.* **7**, 6170 (2017).
46. M. A. Milani, D. González, R. Quijada, N. R. S. Basso, M. L. Cerrada, D. S. Azambuja and G. B. Galland, *Compos. Sci. Technol.* **84**, 1 (2013).
47. R. Tankhiwale and S. K. Bajpai, *Colloids Surf. B Biointerf.* **90**, 16 (2012).
48. P. J. P. Espitia, N. de, F. F. Soares, J. S. dos Reis Coimbra, N. J. de Andrade, R. S. Cruz and E. A. A. Medeiros, *Food Bioprocess Technol.* **5**, 1447 (2012).
49. N. Lepot, M. K. Van Bael, H. Van den Rul, J. D'Haen, R. Peeters, D. Franco and J. Mullens, *J. Appl. Polym. Sci.* **120**, 1616 (2011).
50. A. Marra, C. Silvestre, D. Duraccio and S. Cimmino, *Int. J. Biol. Macromol.* **88**, 254 (2016).
51. J.-E. An, G. W. Jeon and Y. G. Jeong, *Fibers Polym.* **13**, 507 (2012).
52. M. El Achaby, F. Arrakhiz, S. Vaudreuil, A. el Kacem Qaiss, M. Bousmina and O. Fassi-Fehri, *Polym. Compos.* **33**, 733 (2012).
53. J.-Z. Xu, C. Chen, Y. Wang, H. Tang, Z.-M. Li and B. S. Hsiao, *Macromolecules* **44**, 2808 (2011).
54. K. Kalaitzidou, H. Fukushima, P. Askeland and L. T. Drzal, *J. Mater. Sci.* **43**, 2895 (2008).
55. Z. H. Zhang, W. X. Zhou, Y. D. Li and C. M. Yan, *Adv. Mater. Res.* **535–537**, 243 (2012).
56. G. J. Mao and P. Wang, *Mater. Sci. Technol.* **17**, 422 (2009).
57. Y. Yuan, Y. Wang and B. Xie, Preparation and properties of polyimide/SiO₂ hollow spheres composite films with good dielectric property and amazing XRD spectra, in *Material Science Forum* (Trans Tech Publications, 2011), pp. 584–587.
58. C. S. Wu and C. H. Tsou, *J. Polym. Res.* **26**, 44 (2019).
59. C.-H. Tsou, W.-S. Hung, C.-S. Wu, J.-C. Chen, C.-Y. Huang, S.-H. Chiu, C.-Y. Tsou, W.-H. Yao, S.-M. Lin, C.-K. Chu, C.-C. Hu, K.-R. Lee and M.-C. Suen, *Mater. Sci.* **20**, 446 (2014).
60. C.-Y. Tsou, C.-L. Wu, C.-H. Tsou, S.-H. Chiu, M.-C. Suen and W.-S. Hung, *Polym. Sci. Ser. B* **57**, 473 (2015).
61. Y. Yao, M. R. De Guzman, H. Duan, C. Gao, X. Lin, Y. Wen, J. Du, L. Lin, J.-C. Chen, C.-S. Wu, M.-C. Suen, Y. Sun, W.-S. Hung and C.-H. Tsou, *Chinese J. Polym. Sci.* **38**, 898 (2020).
62. J. Rhim, S. Lee and S. Hong, *J. Food Sci.* **76**, N40 (2011).
63. G. Zehetmeyer, R. M. D. Soares, A. Brandelli, R. S. Mauler and R. V. B. Oliveira, *Polym. Bull.* **68**, 2199 (2012).
64. C. Li, J. Zha, H. Long, S. Wang, D. Zhang and Z. Dang, *Compos. Sci. Technol.* **153**, 111 (2017).
65. A. Vassiliou, D. N. Bikiaris, K. Chrissafis, K. M. Paraskevopoulos, S. Y. Stavrev and A. Docoslis, *Compos. Sci. Technol.* **68**, 933 (2008).
66. S. Lei, P. Ren, L. Xu, H. Wang and H. Cui, *Polym. Mater. Sci. Eng.* **31**, 138 (2015).

67. P. Atayev and M. Oner, *J. Plast. Film Sheeting*, **30**, 248 (2014).
68. M. Pannirselvam, A. Genovese, M. Jollands, S. N. Bhattacharya and R. A. Shanks, *Exp. Polym. Lett.*, **2**, 429 (2008).

Supplementary Materials

Table S1. Energy-dispersive X-ray spectroscopic data for G-nZnO.

G-nZnO	unn. C ^a (wt.%)	norm. C ^b (wt.%)	Atom. C ^c (at.%)	C Error (1 Sigma) ^d (wt.%)
C	55.81	41.94	59.84	8.24
Zn	31.13	23.39	6.13	0.92
O	39.12	29.40	31.48	5.38
Cl	7.02	5.27	2.55	0.27

Note: ^aUnnormalized concentration, ^bNormalized concentration, ^cAtomic concentration and ^dError in concentration at sigma level.

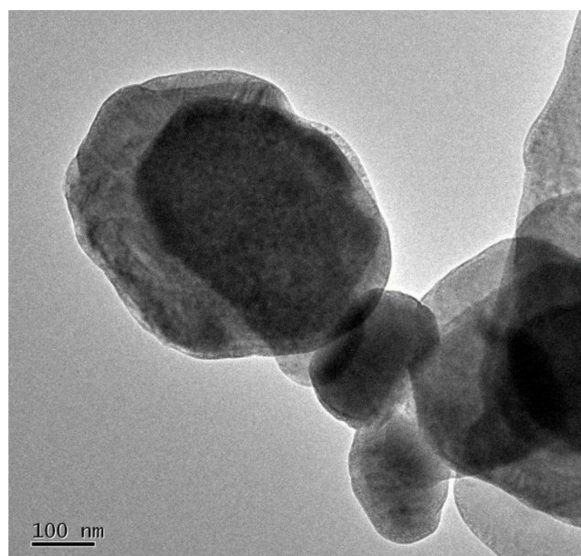


Fig. S1. TEM image of nZnO.

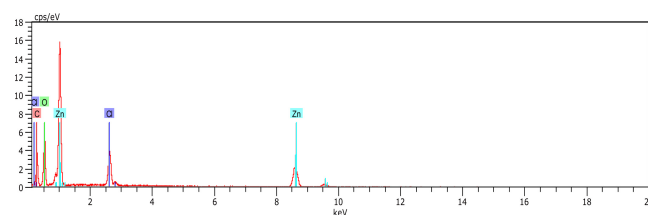


Fig. S2. Energy-dispersive X-ray spectroscopic analysis for G-nZnO.

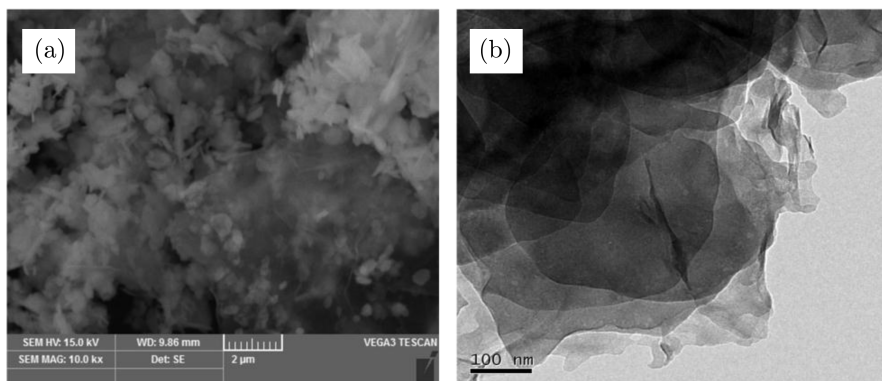


Fig. S3. The morphology of G-nZnO: (a) SEM and (b) TEM.

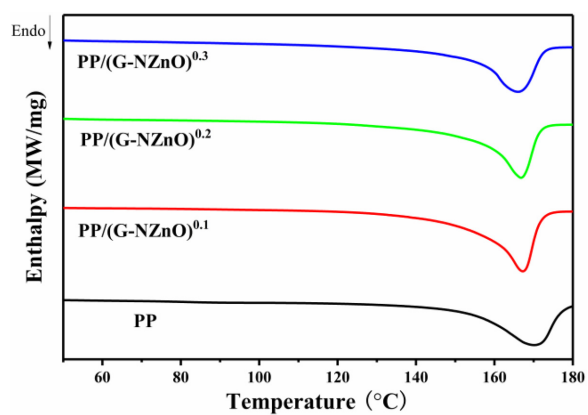


Fig. S4. Differential scanning calorimetry for PP and its nanocomposites.